Cybersecurity - Undergraduate Level Assessment 15 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. A _______________ is a sequential segment of the memory location that is allocated for containing some data such as a character string or an array of integers.
   (a) stack
   (b) queue
   (c) external storage
   (d) buffer
2. Old operating systems like ___________ and NT-based systems have buffer-overflow attack a common vulnerability.
   (a) Windows 7
   (b) Chrome
   (c) IOS 12
   (d) UNIX
3. Which of the following are most vulnerable to injection attacks?
   (a) Session IDs
   (b) Registry keys
   (c) Network communications
   (d) SQL queries based on user input
4. In a ____________________ attack, the extra data that holds some specific instructions in the memory for actions is projected by a cyber-criminal or penetration tester to crack the system.
   (a) Phishing
   (b) MiTM
   (c) Buffer-overflow
   (d) Clickjacking
5. A _________________ is a method in which a computer security mechanism is bypassed untraceable for accessing the computer or its information.
   (a) front-door
   (b) backdoor
   (c) clickjacking
   (d) key-logging

True / False (10 marks)

Answer True or False

6. True or False: Integrity is often regarded as a primary security property alongside confidentiality and availability.
7. True or False: A proxy firewall filters traffic at the application layer, examining data packets for content and context.
8. True or False: Troya is classified as a remote Trojan, designed to compromise systems from a distance.
9. True or False: Message confidentiality guarantees that the data will remain unchanged during transmission, regardless of the sender's intent.
10. True or False: Authentication is a process that verifies the identity of users and is not considered a security exploit.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a python function to check whether the count of divisors is even or odd.
12. Write a python function to count the maximum number of equilateral triangles that can be formed within a given equilateral triangle.

End of Paper